

Subject Description Form

Subject Code	AP50001
Subject Title	Radiation Physics
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	Nil
Objectives	This subject will teach the students the fundamental physics and practices of radiation, medical imaging and nuclear medicines, including basics of classical physics and modern physics, ionizing and non-ionizing radiations, physical quantities and units of radiation, atomic and nuclear structures, basic nuclear and atomic physics, interaction of photons and particles with matter, and radioactivity.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Understand the basic properties of radiation; - Describe the structure of the atom and explain the fundamental intra-atomic phenomena; - Understand the detection mechanisms and quantification of radiation; - Explain the processes in the production of X-rays and the factors which influence them; - Describe and analyze the interaction processes of different types of ionizing radiations with matter and explain their roles in medical imaging and radiation therapy; - Demonstrate understanding of the nuclear decays and their physical origin;
Subject Synopsis/ Indicative Syllabus	Basic Radiation Physics ■ Fundamental and derived physical quantities ■ Basic laws and theories of classical and modern physics ■ Mass-energy equivalence and wave-particle duality ■ Classification of radiations ■ Ionizing and non-ionizing radiations ■ Detection of radiations ■ Physical quantities and units of radiations Atomic Structure ■ Basic atomic models, including Thomson model, Rutherford model, Bohr models of hydrogen atom and multi-electron atom ■ Quantum model of atom ■ Electron transitions ■ Simple band theory

	<u>Assignments:</u> There are 5 assignments in this subject. They evaluate students' understanding of the subject. <u>Short tests:</u> There are 2 short tests. There are to evaluate the students' understanding. <u>Written examination:</u> There is one written examination. It provides an objective and integrative means to assess student's understanding and analytic ability in the subject areas.	
Student Study Effort Expected	Class contact:	
	■ Lecture	33 Hrs.
	■ Tutorial	6 Hrs.
	Other student study effort:	
	■ Self Study	63 Hrs.
	■ Site visit	6 Hrs.
	■ Total student study effort	108 Hrs.
Reading List and References	D. T. Graham. <i>Principles and Applications of Radiological Physics</i>, Churchill Livingstone; 6th edition (July 27, 2012). T. E. Johnson. <i>Introduction to Health Physics</i>, McGraw-Hill Education; 5th Edition (June 5, 2017). J. J. Bevelacqua. <i>Health Physics: Radiation-Generating Devices, Characteristics, and Hazards</i>, Wiley-VCH; 1 edition (July 12, 2016). T. Kuppusamy. <i>Basic Radiological Physics</i>, Jaypee Brothers Medical Publishers; 2nd edition (July 17, 2017). A. Adler. <i>Introduction to Radiologic and Imaging Sciences and Patient Care</i>, Saunders; 7th edition (November 12, 2018). G. F. Knoll. <i>Radiation Detection and Measurement</i>, Wiley India; 4th edition (Jan 01, 2017). N. Tsoulfanidis. <i>Measurement and Detection of Radiation</i>, CRC Press; 4th edition (April 24, 2015). J. Johnston. <i>Essentials of Radiographic Physics and Imaging</i>, Mosby; 2nd Edition (November 18, 2015). N. B. Smith. <i>Introduction to Medical Imaging: Physics, Engineering and Clinical Applications</i>, Cambridge University Press; 1st edition (December 15, 2010). R. R. Carlton. <i>Radiographic Imaging Concepts and Principles</i>, Cengage Learning (2013). R. R. Carlton. <i>Principles of Radiographic Imaging: An Art and A Science</i>, Cengage Learning; 6th edition (January 3, 2019).	